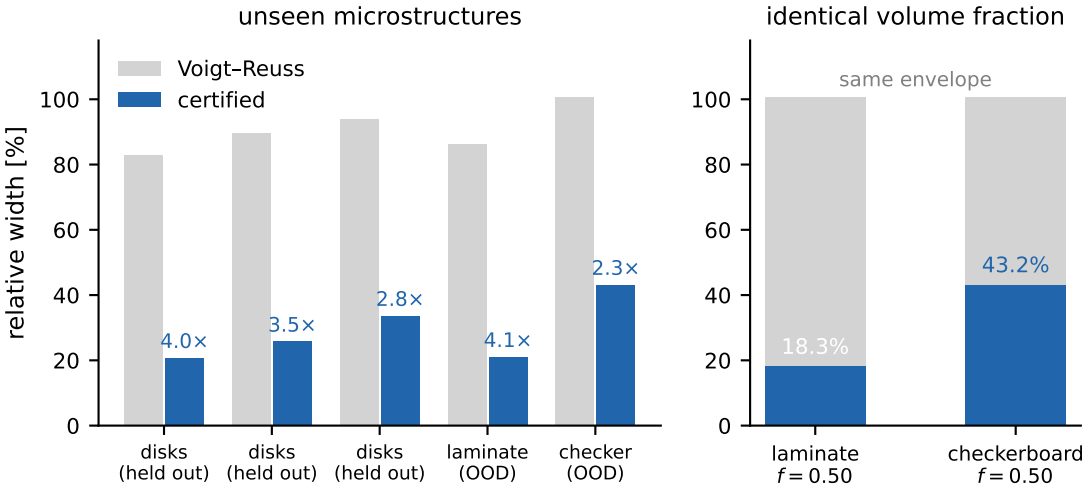
unseen microstructures
identical volume fraction
relative width [%]
Voigt–Reuss
certified
same envelope
4.0×
3.5×
2.8×
4.1×
2.3×
18.3%
43.2%
0
20
40
60
80
100
disks (held out)
disks (held out)
disks (held out)
laminate (OOD)
checker (OOD)
laminate $f = 0.50$
checkerboard $f = 0.50$